

INFRASTRUCTURE PORTFOLIO

HomeLab

DevOps Infrastructure & AI/ML Experimentation Platform

3-Node Proxmox Cluster

HashiCorp Nomad Orchestration

Hybrid AI/ML Infrastructure

PROJECT OVERVIEW

MISSION

Build a production-grade infrastructure platform for learning modern DevOps practices, container orchestration, and AI/ML experimentation through hands-on implementation.

APPROACH

Systematic, phase-based development

Operational maturity before automation

Documentation-first methodology

FOCUS AREAS

Container orchestration with Nomad

Service discovery and observability

AI/ML workload deployment

VALUES

Practical over perfect

Learn by doing

Incremental progress

INFRASTRUCTURE ARCHITECTURE

COMPUTE LAYER

3x Proxmox Hosts

Intel-based bare metal servers

Virtual Machine Cluster

Flexible workload distribution

Mac Studio (M1 Max)

GPU-accelerated AI inference

ORCHESTRATION

HashiCorp Nomad

Container orchestration

Consul

Service discovery

Traefik

Reverse proxy

STORAGE

Synology NAS (10TB)

NFS Shared Volumes

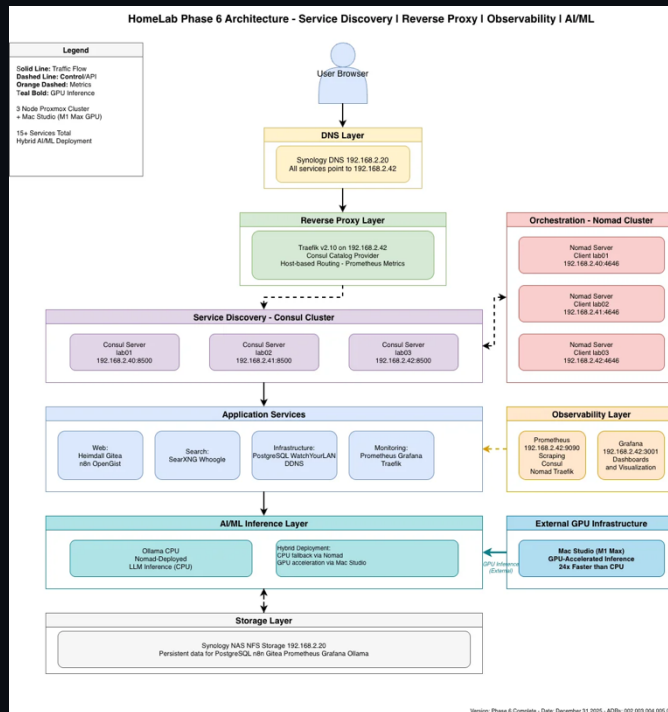
OBSERVABILITY

Prometheus + Grafana

Pushover Alerting

Designed for operational simplicity and learning value

SYSTEM ARCHITECTURE



Layered architecture: DNS → Proxy → Service Discovery → Orchestration → Services → AI/ML → Storage

TECHNOLOGY STACK

CONTAINER PLATFORM

Docker runtime across all nodes

LinuxServer.io image ecosystem

Privileged mode for complex services

SERVICE MESH

Consul service registry

Automated health checks

Dynamic service discovery

AI/ML INFRASTRUCTURE

Ollama for local LLM inference

Hybrid GPU/CPU deployment

24x performance on M1 Max GPU

MONITORING & ALERTING

Prometheus metrics scraping

Custom + community dashboards

Real-time mobile notifications

Selected for operational simplicity and production-readiness

IMPLEMENTATION JOURNEY

PHASE 1-2 ✓

FOUNDATION

Hardware inventory
documented

Nomad cluster
established

Architecture decisions
recorded

PHASE 3-4 ✓

CORE SERVICES

Consul service discovery

Traefik reverse proxy

Container deployment
patterns

PHASE 5 ✓

OBSERVABILITY

Prometheus metrics stack

Grafana dashboards

Mobile alerting system

PHASE 6 ✓

AI/ML PLATFORM

Hybrid Ollama
deployment

GPU-accelerated
inference

Orchestrated CPU
fallback

Progressive buildout with validation at each milestone

THE HYBRID DECISION

Architecture Question: Where Should Ollama Run?

Core Principle

"Learn by doing (CPU), perform in production (GPU)"

Learning Achieved

Nomad AI orchestration mastered

Performance Delivered

24x speedup when needed

Resilience Bonus

Built-in redundancy

Decision: Both endpoints operational, choose based on use case

KEY LEARNINGS

OPERATIONAL MATURITY FIRST

Automate the running system before automating infrastructure provisioning. Manual operations reveal pain points and inform better automation decisions.

PARALLEL ACCESS

Maintain dual access paths during migrations to ensure service availability and enable safe rollback.

HYBRID ARCHITECTURES

Balance learning value with performance by combining CPU and GPU resources strategically.

DOCUMENTATION TIMING

Document decisions while context is fresh rather than retroactively to capture reasoning and alternatives.

FOCUSED SESSIONS

Systematic session-based work produces more reliable results than marathon implementation efforts.

TECHNICAL ACHIEVEMENTS

24x faster

GPU ACCELERATION

M1 Max GPU vs CPU inference performance

100% uptime

SERVICE AVAILABILITY

Comprehensive health checks and monitoring

3-Node

CLUSTER SCALE

Distributed workload orchestration

PRODUCTION CAPABILITIES

- ✓ Automated service discovery
- ✓ Dynamic load balancing
- ✓ Container health monitoring
- ✓ Mobile alerting integration
- ✓ Custom observability dashboards
- ✓ NFS-backed persistent storage
- ✓ Hybrid AI/ML infrastructure
- ✓ Version-controlled configs
- ✓ Architecture decision records

Enterprise-grade capabilities in a homelab environment

DEPLOYED SERVICES

ORCHESTRATION

Nomad (3 servers, N workers)

Consul service mesh

Traefik reverse proxy

MONITORING

Prometheus metrics

Grafana visualization

Pushover alerting

AI/ML

Ollama (Mac Studio GPU)

Ollama (Nomad CPU)

Local LLM inference

STORAGE

NFS mounts from Synology

Persistent volume support

Shared data access

APPLICATIONS

Containerized workloads

LinuxServer.io images

Custom deployments

NETWORKING

Service-to-service discovery

Automated routing

Health-based failover

All services registered in Consul with comprehensive health checks and observability coverage

Production-ready services with automated management

DEVELOPMENT METHODOLOGY

WORKFLOW PATTERN

Focused work sessions

Real-time documentation

Formal commit process

DOCUMENTATION

Google Drive for strategy

Local Git for configs

Architecture Decision Records

VERSION CONTROL

Git for configurations

Detailed commit messages

DEPLOYMENT

Dry-run validation

Incremental rollout

FUTURE ROADMAP

NEXT PHASE: INFRASTRUCTURE AUTOMATION

Automate infrastructure provisioning and configuration management now that operational patterns are proven and well-understood through manual management.

AUTOMATION TARGETS

Terraform for infrastructure
Ansible for configuration
GitOps workflows

AI/ML EXPANSION

RAG pipeline development
Agent framework exploration
Model fine-tuning experiments

CAPABILITIES

Advanced networking patterns
Security hardening
Disaster recovery testing

Continue learning through building - expand capabilities based on real needs

Sustainable growth path with continued focus on practical learning